

Business Problem

The bookstore industry processes thousands of book sales transactions every day. Each transaction generates valuable business data, including book details, customer information, order quantity, sales amount, and inventory levels. However, when this information is stored in raw database tables, it becomes difficult for business managers to monitor sales performance, understand customer purchasing behavior, manage inventory efficiently, and make informed business decisions.

The goal of this analysis is to:

- Analyze total sales revenue generated from book orders.
- Identify the most and least ordered books.
- Monitor inventory levels and remaining stock.
- Analyze book sales by genre and author.
- Identify top spending customers.
- Measure customer purchasing frequency.
- Evaluate average book prices across different genres.
- Support management with data-driven insights for inventory planning, sales optimization, and customer retention.

Dataset Overview

The dataset contains transactional and master data representing a bookstore's sales operations. It captures information related to books, customers, and sales orders for business analysis.

Dataset Includes

Books

- Book_ID
- Title
- Author
- Genre
- Published_Year

- Price
- Stock

Customers

- Customer_ID
- Name
- Email
- Phone
- City
- Country

Orders

- Order_ID
- Customer_ID
- Book_ID
- Order_Date
- Quantity
- Total_Amount

Project KPIs

- Total Books Available
- Total Customers
- Total Orders
- Total Revenue
- Average Book Price
- Total Books Sold
- Remaining Inventory
- Best Selling Book
- Top Spending Customer

- Most Popular Genre
- Author-wise Sales
- Repeat Customers

Summary Statistics

Key Business Metrics

Sales Performance

- The bookstore generates revenue through customer book purchases.
- Certain books receive significantly higher order volumes than others.
- Multiple customers place repeat orders, indicating customer loyalty.
- Revenue is primarily driven by high-demand books and premium-priced titles.

Genre Performance

- Fantasy and Fiction genres contribute a significant portion of book sales.
- Genre-wise analysis helps identify customer reading preferences.
- Low-performing genres provide opportunities for promotional campaigns.

Customer Purchasing Behavior

- Some customers place multiple orders, demonstrating repeat purchasing behavior.
- Top spending customers contribute a larger share of total revenue.
- Customer purchase frequency supports loyalty program planning.

Inventory Analysis

- Remaining stock is calculated after fulfilling customer orders.
- Low-stock books can be identified for timely replenishment.
- Inventory monitoring reduces the risk of stock shortages.

Data Validation

To ensure accurate reporting and reliable business insights, the following validation checks were performed:

- Removed duplicate records.
- Verified unique Book_ID, Customer_ID, and Order_ID values.
- Checked for missing or NULL values.
- Ensured stock values were greater than or equal to zero.
- Validated book prices to prevent invalid values.
- Verified foreign key relationships between Books, Customers, and Orders.
- Checked order quantities for valid positive values.
- Validated total order amounts.
- Ensured referential integrity across all tables.

These validation steps improved data quality and ensured that SQL query results accurately represent bookstore operations.

Research Questions & Key Findings

1. Which books are ordered the most?

Finding

Certain books receive significantly higher order frequencies, indicating strong customer demand.

Business Impact

- Increase inventory for best-selling books.
- Prevent stock shortages.
- Improve customer satisfaction through product availability.

2. Which genre contributes the highest book sales?

Finding

Genre-wise analysis identifies the categories generating the highest sales volume.

Business Impact

- Focus marketing campaigns on high-performing genres.

- Optimize inventory based on customer preferences.
- Expand popular book categories.

3. Who are the highest spending customers?

Finding

A small group of customers contributes the highest total purchase value.

Business Impact

- Develop loyalty reward programs.
- Offer exclusive discounts.
- Improve customer retention.

4. Which books require inventory replenishment?

Finding

Books with the lowest remaining stock are identified after fulfilling customer orders.

Business Impact

- Prevent stockouts.
- Improve inventory planning.
- Reduce lost sales opportunities.

5. Which authors generate the highest sales?

Finding

Author-wise sales analysis identifies authors contributing the highest book sales.

Business Impact

- Promote popular authors.
- Increase inventory for high-demand authors.
- Support future purchasing decisions.

6. Which customers place repeat orders?

Finding

Several customers have placed two or more orders.

Business Impact

- Identify loyal customers.
- Launch personalized marketing campaigns.
- Increase customer lifetime value.

7. What are the overall business insights?

Key Findings

- The bookstore successfully tracks books, customers, and sales transactions using a relational database.
- Total revenue is generated through customer book purchases.
- Best-selling books contribute significantly to overall sales performance.
- Genre-wise analysis reveals customer reading preferences.
- Repeat customers indicate strong customer engagement.
- Inventory analysis helps identify books requiring replenishment.
- Author-wise sales support strategic purchasing decisions.
- SQL-based analysis enables data-driven decision-making for inventory management, customer retention, and sales optimization.